Semantic Alignment, Chronological Distance, and Citation-Network Prominence in Citation Formation: Evidence from Ten Citation Corpora

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Abstract

What survives after temporal leakage is rigorously removed from citation link prediction? This paper answers that question across ten large-scale bibliographic corpora. We introduce a strongly leakage-mitigated methodological framework in which all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We formalize four forms of temporal leakage—future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and apply this framework to ten bibliographic datasets retrieved from Dimensions.ai, spanning five broad scholarly domains: Science (Protein Folding, CRISPR), Engineering (Additive Manufacturing, Corrosion Protection), BioMed (Neuroblastoma, Osteosarcoma), Social Science (Income Inequality, Organizational Behavior), and Humanities (Film Studies, Memory Studies). Each corpus covers the period 1975–2024. Gradient Boosting achieves ROC-AUC scores ranging from 0.530 to 0.734 under temporally rigorous evaluation. The goal of this study is not to maximize citation prediction accuracy, but to decompose citation formation into distinct explanatory components under temporally rigorous conditions. We decompose citation formation into three components: semantic alignment (S), chronological distance (T), and citation-network prominence factors (N_{prom}). Under this decomposition, chronological recency (T) dominates prediction across almost all corpora. Once the candidate set is restricted to plausible contemporaneous papers from the same topical corpus, directional semantic alignment adds little discriminative power beyond chronological recency: in four of the ten corpora, cited papers exhibit marginally lower directional semantic alignment to the citing paper than non-cited candidates. When citations occur despite weak semantic alignment, they tend to be temporally proximate, though they do not exhibit a consistent prominence pattern. We conclude that under leakage-controlled conditions, citation formation exhibits substantial domain-specific heterogeneity, but chronological distance factors consistently contribute more predictive signal than directional semantic alignment or citation-network prominence factors.

Keywords: citation link prediction; temporal leakage; S-T-N decomposition; bibliometric machine learning; SBERT semantic alignment; gradient boosting; feature ablation; SHAP; scholarly domain comparison; Dimensions.ai; temporally capped graph; citation-network prominence

1 Introduction

Scientific citations constitute the primary mechanism through which scholarly knowledge is accumulated, validated, and integrated into the scientific record. Citation networks therefore encode not only the intellectual lineage of individual publications but also the evolving structure of scientific communities and disciplinary attention. Predicting the formation of new links within these networks—citation link prediction—has consequently emerged as a central problem at the intersection of informetrics, network science, information retrieval, and machine learning (Wang et al., 2013; Shibata et al., 2012; He et al., 2010; Beel et al., 2016). Accurate citation prediction has practical applications in literature recommendation systems, research-front detection, and scholarly search, while also contributing to the broader theoretical understanding of how scientific attention is allocated across increasingly large and fragmented bodies of published knowledge.

At the same time, the methodological validity of citation prediction remains deeply sensitive to temporality. Citation networks are cumulative and dynamically evolving systems in which the information available at the time of citation differs fundamentally from the information observable retrospectively. Nevertheless, many machine learning studies construct features from the fully aggregated static graph, thereby allowing future information to influence past predictions implicitly. For example, computing a paper's PageRank on the complete 2024 citation network and using it to predict a citation that occurred in 2010 allows the model to exploit information about the paper's eventual scientific success that was unavailable to the citing author at the time of writing. Such future topology leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that actually governed citation behavior. Temporal leakage has become an increasingly important concern in applied machine learning more broadly, particularly in temporally ordered prediction tasks (Fortunato et al., 2018), yet citation prediction remains especially vulnerable because the chronological distance structure of bibliometric data is frequently ignored in benchmark construction and feature engineering.

Accordingly, the present study is positioned not primarily as a new recommendation architecture, but as a strongly leakage-mitigated evaluation framework for bibliometric citation prediction. The goal of this study is not to maximize citation prediction accuracy, but to decompose citation formation into distinct explanatory components under temporally rigorous conditions. Our primary contribution is methodological. We introduce a formal taxonomy of temporal leakage and provide explicit mathematical definitions of the temporally capped graph, the constrained candidate space, and the ranking objective. We engineer a feature set encompassing structural overlap, citation-network prominence, chronological distance dynamics, and semantic alignment, while ensuring that all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We pair this feature construction with a temporally constrained evaluation protocol incorporating hard negative sampling, rolling chronological distance windows, and top-k ranking experiments designed to establish a rigorous intermediate step between pairwise discrimination and large-scale retrieval evaluation.

Our secondary contribution is the application of this framework to ten topical citation graphs retrieved from Dimensions.ai, spanning Science, Engineering, BioMed, Social Science, and Humanities. By applying identical feature engineering and evaluation protocols across these diverse scholarly corpora, we decompose citation formation into three components: semantic alignment (S), chronological distance (T), and citation-network prominence factors (N_{prom}). This decomposition directly answers the question: how much of citation formation is explained by semantic proximity, how much by chronological aging, and how much by accumulated scholarly visibility?

Our principal empirical finding is that chronological recency (T) dominates prediction across almost all corpora. Once the candidate set is restricted to plausible contemporaneous papers from the same topical corpus, directional semantic alignment adds little discriminative power beyond chronological recency. In four of the ten corpora, cited papers exhibit marginally lower directional semantic alignment to the citing paper than non-cited candidates. When citations occur despite weak semantic alignment, they tend to be temporally proximate, though they do not exhibit a consistent prominence pattern. We embrace this finding not as a methodological failure but as a substantive scientometric result: under leakage-controlled conditions, citation formation exhibits substantial domain-specific heterogeneity, but chronological distance factors consistently contribute more predictive signal than directional semantic alignment or citation-network prominence factors.

Finally, we explicitly delimit the scope of the present study. Our retrieval experiments approximate recommendation realism through constrained candidate pools rather than full-corpus search; our graph-theoretic features function primarily as baseline enrichment rather than as sophisticated higher-order topological descriptors; and our semantic representations are based on SBERT embeddings whose pre-training corpora introduce a systemic vector of semantic leakage that cannot be perfectly mitigated without chronologically re-trained language models. These limitations are acknowledged explicitly in order to distinguish clearly between the methodological guarantees established here and broader claims that remain outside the scope of the present work.

2 Literature Review

2.1 Citation Dynamics and Network Evolution

The dynamics of scientific citation have long been associated with cumulative advantage processes in which highly cited papers attract future citations at disproportionate rates. Merton formalized this mechanism sociologically as the Matthew Effect (Merton, 1968), while network science later operationalized it mathematically through preferential attachment models (Barabási and Albert, 1999). In citation networks, preferential attachment implies that the probability of acquiring new citations grows as a function of existing citation counts. However, degree accumulation alone cannot fully explain citation behavior because scientific attention is simultaneously shaped by aging, obsolescence, topical relevance, and disciplinary visibility (Price, 1965). Wang, Song, and Barabási (Wang et al., 2013) demonstrated that long-term citation trajectories emerge from the interaction among preferential attachment, intrinsic fitness, and chronological aging. These mechanisms directly motivate the present framework: historical prestige operationalizes cumulative advantage, semantic alignment approximates topical fitness, and chronological distance captures aging and obsolescence effects.

2.2 Social and Semantic Proximity in Citation

Beyond prestige, a growing body of research has demonstrated that social proximity and semantic proximity are powerful predictors of citation behavior, often rivaling or exceeding the effect of prestige for the majority of papers. Newman's foundational work on co-authorship networks established that scientific collaboration is highly clustered and that co-authorship strongly predicts future collaboration and citation (Newman, 2004). Kumar extended this analysis to show that citation probability decays rapidly with social distance in co-authorship networks (Kumar, 2015). Martin et al. demonstrated that the social structure of scientific communities—operationalized through co-authorship, institutional affiliation, and conference participation—is a major determinant of citation patterns (Martin et al., 2013).

The landmark "Citation proximus" study by Kozlowski et al. (Kozlowski et al., 2025) provided rigorous statistical evidence for the joint role of social and semantic proximity using Generalized Linear Mixed Models, demonstrating that these factors together explain a substantial fraction of the variance in citation behavior across multiple scientific fields. While the present study focuses on semantic and structural features due to the computational constraints of constructing historical co-authorship networks across the entire candidate space, we recognize social proximity as a crucial parallel mechanism that operates alongside the topological and textual signals evaluated here.

2.3 Temporal Link Prediction and Leakage

Temporal link prediction—predicting which edges will form in a dynamic network given its history up to time t (Liben-Nowell and Kleinberg, 2007)—is a well-established problem in network science. In citation networks, temporality is not merely methodological but fundamentally causal, since a citation can only occur after the publication of the cited work. Nevertheless, many citation-prediction studies construct features retrospectively from fully aggregated graphs, thereby allowing future information to influence past predictions implicitly. Such temporal leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that governed citation behavior at the time the citation decision was made.

Prior studies have explored structural overlap measures and chronological distance features in citation prediction (Shibata et al., 2012; Chakraborty et al., 2015), but relatively few have formalized leakage explicitly as a methodological problem. More broadly, leakage-aware evaluation has become an increasingly important concern in applied machine learning because temporally ordered systems are especially vulnerable to hidden future-information contamination (Fortunato et al., 2018). These concerns are particularly acute in bibliometric prediction because static citation graphs implicitly encode future scientific prominence. The present study extends this literature by introducing an explicit taxonomy of temporal leakage together with formally defined temporally capped graph representations and rolling chronological distance evaluation procedures.

2.4 Bibliographic Coupling and Structural Proximity

Bibliographic coupling (Kessler, 1963) and co-citation analysis (Small, 1973) remain foundational measures of intellectual proximity in bibliometrics. Bibliographic coupling captures shared intellectual ancestry through common references, whereas co-citation reflects the extent to which subsequent literature jointly associates two papers. Both measures have been widely applied to research-front detection, science mapping, and scholarly clustering.

Within citation prediction, these mechanisms provide structural representations of topical and disciplinary proximity embedded within the evolving citation graph. Structural overlap features derived from shared references and shared citers therefore complement semantic textual similarity by capturing relationships perceived through the existing literature itself. In the present framework, bibliographic coupling and co-citation are incorporated explicitly as temporally constrained structural features whose predictive contribution can be isolated through ablation analysis.

2.5 Machine Learning and Semantic Citation Prediction

Early machine learning approaches to citation prediction relied primarily on local structural features combined with linear classifiers (Liben-Nowell and Kleinberg, 2007). More recent work increasingly incorporates semantic representations derived from titles, abstracts, and full text. Traditional vector-space approaches based on TF-IDF representations remain widely used because of their interpretability and computational transparency, while more recent systems employ contextual neural embeddings such as BERT and SciBERT (Devlin et al., 2019). Content-based citation recommendation has demonstrated that semantic information substantially improves predictive performance beyond topology alone (Bhagavatula et al., 2018). Ensemble learning methods, including Random Forest (Breiman, 2001) and Gradient Boosting (Natekin and Knoll, 2013), have similarly shown strong predictive performance in bibliometric tasks characterized by complex non-linear interactions among prestige, structure, semantics, and temporality. Kozlowski et al. (Kozlowski et al., 2025) further demonstrated that semantic alignment is among the strongest predictors of citation behavior, rivaled primarily by direct social or collaborative ties. The present study builds upon this literature while evaluating semantic and structural signals under explicitly leakage-mitigated temporal constraints across ten disciplines.

2.6 Information Retrieval Evaluation and Learning-to-Rank

Binary classification metrics such as ROC-AUC and F1-score provide only partial evaluation for citation recommendation systems because scholarly recommendation is fundamentally a ranking problem (Manning et al., 2008; Powers, 2011). Information retrieval frameworks developed in large-scale evaluation settings such as TREC (Voorhees and Harman, 2005) therefore emphasize ranking-sensitive measures including Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG@k), and Precision@k (Järvelin and Kekäläinen, 2002). Learning-to-rank architectures increasingly optimize directly for such retrieval-oriented objectives in modern recommendation systems (Liu, 2009).

These ranking-oriented evaluation paradigms are especially important in citation prediction because strong pairwise discrimination does not necessarily translate into realistic retrieval performance over large candidate spaces. Citation recommendation systems have progressively moved toward retrieval-oriented pipelines combining semantic search, ranking models, and large candidate pools (Bhagavatula et al., 2018). At large scale, these systems frequently rely on approximate nearest-neighbor retrieval and multi-stage ranking architectures to ensure computational feasibility. Although the present study does not implement neural retrieval or full-corpus approximate nearest-neighbor search, our top-k ranking evaluation over temporally constrained candidate pools provides a rigorous intermediate framework bridging pairwise discrimination and realistic retrieval-oriented evaluation.

3 Ten Disciplinary Case Studies

3.1 Bibliographic Data Infrastructure

All ten datasets were constructed from the Dimensions database (Hook et al., 2018), a comprehensive, linked research information system integrating publications, citations, grants, patents, and policy documents. Its coverage is among the broadest available, and it has been shown to provide a high degree of completeness and quality in publication metadata (Delgado-Quirós and Ortega, 2024; Nguyen et al., 2022). Programmatic access to the Dimensions API was facilitated via the `dimcli` Python client (Peroni and Shotton, 2021). For each dataset, we queried the Dimensions DSL using keyword searches in title and abstract, restricted to journal articles published 1975–2024.

The use of a single bibliographic infrastructure for all ten datasets ensures that the comparative analysis is not confounded by differences in metadata availability, coverage, or indexing policy across platforms. All ten corpora are therefore directly comparable in terms of data provenance, field availability, and citation-graph construction methodology.

3.2 The Ten Citation Graphs

The ten datasets span five broad domains: Science (Protein Folding, CRISPR), Engineering (Additive Manufacturing, Corrosion Protection), BioMed (Neuroblastoma, Osteosarcoma), Social Science (Income Inequality, Organizational Behavior), and Humanities (Film Studies, Memory Studies). Table 1 summarizes the topological characteristics of the ten citation graphs.

Table 1: Descriptive statistics of the ten topical citation graphs extracted from Dimensions (1975–2024).

Domain	**Dataset**	**Papers**	**Papers in internal citations**	**Internal citations**	**Density**	**GCC size**
Science	Protein Folding	82,428	73,433 (89.1%)	889,384	2.62e-04	72,970
Science	CRISPR	64,484	49,224 (76.3%)	753,711	3.63e-04	48,937
BioMed	Neuroblastoma	57,319	44,290 (77.3%)	326,670	1.99e-04	43,662
BioMed	Osteosarcoma	61,187	46,131 (75.4%)	389,874	2.08e-04	45,365
Engineering	Additive Manuf.	58,541	48,937 (83.6%)	575,853	3.36e-04	48,190
Engineering	Corrosion Prot.	26,486	18,861 (71.2%)	125,884	3.59e-04	18,112
Social Science	Income Inequality	40,279	26,974 (67.0%)	136,433	1.68e-04	26,189
Social Science	Org. Behavior	75,622	42,269 (55.9%)	242,381	8.48e-05	39,904
Humanities	Film Studies	393,342	280,525 (71.3%)	903,486	1.17e-05	264,167
Humanities	Memory Studies	347,195	247,507 (71.3%)	1,820,834	3.02e-05	237,708

The datasets exhibit expected variations in citation density and clustering across domains. Science corpora (Protein Folding, CRISPR) and Engineering corpora (Additive Manufacturing, Corrosion Protection) display the highest citation densities (2.62e-04 to 3.63e-04), reflecting the rapid and internally coherent citation cultures of these fields. BioMed corpora (Neuroblastoma, Osteosarcoma) are moderately dense. Social Science corpora (Income Inequality, Organizational

Behavior) are substantially sparser, reflecting citation cultures that draw from broader literatures. The Humanities corpora (Film Studies, Memory Studies) have the largest paper counts (393,342 and 347,195 respectively) but the lowest citation densities (1.17e-05 and 3.02e-05), consistent with the monograph-heavy and cross-disciplinary citation practices of these fields.

4 Methodology

4.1 Mathematical Formalization

We define a citation network as a directed temporally annotated graph $G = (V, E, T)$, where each node $v \in V$ represents a scientific paper with publication timestamp t_v, and each directed edge $(u, v) \in E$ represents a citation from paper u to paper v. The network is temporally acyclic, i.e.,

$$(u, v) \in E \implies t_u > t_v.$$

Temporally Capped Graph. For a citing paper A published at time t_A, we define the temporally capped graph

$$G_{t_A-1} = (V_{<t_A}, E_{<t_A}),$$

where

$$V_{<t_A} = \{v \in V \mid t_v < t_A\}$$

and

$$E_{<t_A} = \{(u, v) \in E \mid t_u < t_A \text{ and } t_v < t_A\}.$$

All features associated with the candidate pair (A, B) are computed exclusively from G_{t_A-1}.

Supervised Learning Objective. Given a candidate pair (A, B), define

$$y(A, B) = \begin{cases} 1 & \text{if } (A \to B) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The supervised learning task is to estimate

$$P(y = 1 \mid \mathbf{x}_{AB}),$$

where $\mathbf{x}_{AB}$ denotes the feature vector associated with the pair (A, B).

Constrained Candidate Space. For a citing paper A and a true cited paper B, the constrained candidate space is defined as

$$C(A) = \{B' \in V_{<t_A} \mid |t_{B'} - t_B| \leq 3,\ \text{topic}(B') = \text{topic}(B),\ (A \to B') \notin E\}.$$

Ranking Objective. Given a query paper A and a candidate pool

$$\{B\} \cup \{B'_1, \ldots, B'_{k-1}\}$$

drawn from $C(A)$, the ranking objective is to assign a score $s(A, \cdot)$ to each candidate such that the true cited paper B is ranked highest. The scoring function $s(A, B)$ is defined as the posterior class-probability estimate returned by the Gradient Boosting classifier.

Ranking quality is evaluated using Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG@k):

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\mathrm{rank}_q(B_q)},$$

$$\mathrm{NDCG@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathrm{DCG@}k(q)}{\mathrm{IDCG@}k(q)},$$

where Q denotes the set of query papers and $\mathrm{rank}_q(B_q)$ is the rank assigned to the true cited paper B_q for query q.

4.2 A Taxonomy of Temporal Leakage

We define four distinct forms of temporal leakage that commonly affect citation prediction systems.

1. **Future Citation Leakage.** The target citation link ($A \to B$) is present in the graph used for feature computation. This is eliminated by removing the target edge prior to feature construction.

2. **Future Topology Leakage.** Network centrality or overlap measures are computed using nodes and edges entering the network at or after t_A. This allows the model to exploit the future scientific prominence of the cited paper. We eliminate this by restricting all structural computations to $G_{t_A - 1}$.

3. **Metadata Leakage.** Retrospectively updated metadata are used during prediction. For example, using a journal's 2024 Impact Factor to predict a citation occurring in 2010 introduces future information unavailable at the time of citation. We mitigate this by using only historically available metadata.

4. **Semantic Leakage.** Text embeddings derived from pretrained language models may implicitly encode citation contexts if the target papers were included in the pretraining corpus. We employ Sentence-BERT (`all-MiniLM-L6-v2`) to compute semantic alignment from the title and abstract fields available in Dimensions. We acknowledge that the static pre-training corpus of SBERT introduces a systemic vector of semantic leakage that cannot be mitigated without chronologically re-trained language models. A fully leakage-free semantic representation would require rolling vocabulary construction or temporally constrained re-training, which we identify as future work.

4.3 Data Collection

All ten datasets were retrieved from Dimensions.ai (Hook et al., 2018) using the Dimensions DSL, querying by keyword in title and abstract and restricting to journal articles published 1975–2024. Representative queries used topical keyword combinations such as `"protein folding"` for the Science–Protein Folding corpus, `"CRISPR"` for the Science–CRISPR corpus, and analogous terms for the remaining eight domains. For each corpus, all records with missing publication years or reference lists were removed after retrieval. The ten datasets collectively span 1975–2024, with record counts ranging from 26,486 (Corrosion Protection) to 393,342 (Film Studies). Abstract availability is high across all corpora. See `outputs/tables/table1_corpus_overview.csv` in the repository for the complete corpus construction details.

4.4 Temporally and Topically Constrained Negatives

Standard link-prediction studies often sample negative edges uniformly at random. In citation networks, this procedure produces trivially separable negatives, such as pairing a contemporary article with an unrelated paper from another discipline or era. We instead employ temporally constrained negatives derived from the candidate space $C(A)$. For each positive pair $(A \to B)$, we sample a non-cited paper B' published within ± 3 years of B and belonging to the same topical corpus. This forces the model to discriminate between contemporaneously available papers, thereby isolating the actual mechanisms influencing citation choice.

The construction of $C(A)$ proceeds in four sequential steps, each relying exclusively on historically available metadata. First, all papers published strictly before t_A are identified from $V_{<t_A}$. Second, this set is filtered to retain only papers whose publication year falls within three years of the true cited paper B. Third, the resulting set is further restricted to papers belonging to the same keyword-defined corpus as B. Fourth, papers already cited by A are excluded. No global network metrics are used at any stage of this filtering process.

This procedure yielded balanced positive-negative pair counts that vary across domains, reflecting the variation in internal citation density documented in Table 1. The resulting pair counts range from 112 (Protein Folding) to 754 (Film Studies). The relatively small number of valid pairs reflects the sparsity of internal citation networks in narrowly defined topical corpora rather than an arbitrary sampling decision. The final prediction set is balanced (1:1 positive-to-negative ratio) and constitutes the exhaustive set of valid pairs under this protocol; it is not a subsample of a larger pool. Specifically, the small counts arise from three compounding constraints: (1) only citations between papers within the same corpus are used as positives, since many papers cite works outside the corpus that are unavailable for evaluation; (2) strict chronological distance ordering ($t_B < t_A$) is enforced, dropping pairs with missing or equal years; and (3) the hard-negative ± 3-year window may contain very few non-cited candidates in sparse corpora, causing some negatives to be skipped. Results for the sparsest corpora should be interpreted with appropriate caution.

4.5 Temporally Aware Feature Engineering

For each candidate pair (A, B), where A is published in year t_A, all features are computed exclusively from the graph $G_{t_A - 1}$.

For the purposes of the S-T-N$_{prom}$ decomposition, the features are grouped into three explanatory blocks. The **Semantic block (S)** contains directional semantic alignment. The **Temporal block (T)** contains the citation time gap. The **Citation-Network Prominence block (N$_{prom}$)** contains chronological distance in-degree and year-capped PageRank, representing the accumulated scholarly visibility and historical citation prominence of the cited paper prior to the citing paper's publication.

Citation-Network Prominence (N$_{prom}$): Temporal In-degree. The in-degree of B in $G_{t_A - 1}$, representing the number of citations received by B prior to t_A.

Citation-Network Prominence (N_{prom}): Year-Capped PageRank. The PageRank score of B computed on G_{t_A-1}. In our decomposition analysis, PageRank provides only modest improvements beyond simple prestige measures; accordingly, we interpret them as operationalizations of accumulated scholarly visibility rather than as higher-order topological descriptors.

Temporal Gap (T). The quantity

$$t_A - t_B,$$

capturing citation aging effects consistent with the chronological distance dynamics described by Wang et al. (2013). A positive value of T indicates that the cited paper is older than the citing paper; a larger T therefore corresponds to an older cited paper.

Common References. The quantity

$$|R_A \cap R_B|,$$

where R denotes the reference list of a paper, operationalizing bibliographic coupling (Kessler, 1963).

Jaccard References. The normalized bibliographic-coupling measure

$$\frac{|R_A \cap R_B|}{|R_A \cup R_B|}.$$

Co-citation Overlap. The number of papers in G_{t_A-1} citing both A and B, operationalizing co-citation (Small, 1973). Because A is newly published at time t_A, this quantity is typically zero unless A previously existed as a preprint. Accordingly, co-citation is retained as a structural baseline but contributes negligible predictive signal, as confirmed by the SHAP analysis.

Semantic Similarity (S): Directional Semantic Similarity. The Kotlerman-Dagan Average Precision directional semantic alignment between the concatenated title and abstract of A and B, computed using SBERT (`all-MiniLM-L6-v2`) embeddings applied to the title and abstract fields available in Dimensions. Higher values indicate greater directional semantic alignment. This representation captures semantic alignment between the intellectual content of the two papers. Abstract availability in Dimensions ranges from 69.3% to 96.6% across the ten datasets; for pairs where one or both abstracts are missing, the similarity is computed from the title alone.

Network-derived quantities for newly published candidate papers present no computational pathology because the temporally capped graph G_{t_A-1} excludes the citing paper A by construction, ensuring that all structural features are computed from the pre-existing network state.

4.6 Model Training and Evaluation

We trained four classifiers: Logistic Regression, Linear SVM, Random Forest, and Gradient Boosting. All models were evaluated using 5-fold stratified cross-validation. To assess chronological distance robustness, we additionally evaluated model performance using a strict chronological distance hold-out split (training on papers published ≤ 2015, testing on papers published $2016 - 2020$). For all ten datasets, we additionally evaluated performance across four rolling chronological distance windows. For ranking evaluation, we constructed top-k candidate pools with

$$k \in \{10, 50, 100\},$$

and computed MRR, NDCG@k, and Precision@k. To quantify uncertainty in the ranking metrics, we estimated 95% bootstrap confidence intervals using 1000 resamples.

To isolate the contribution of each explanatory block, we trained seven sub-models using all combinations of S, T, and N_{prom}: S alone, T alone, N_{prom} alone, S+T, S+N_{prom}, T+N_{prom}, and S+T+N_{prom} (the full model). All seven sub-models used the same 5-fold stratified cross-validation splits and the same Gradient Boosting hyperparameters, ensuring that AUC differences reflect feature contributions rather than evaluation artifacts. We computed three marginal contribution deltas:

$$\Delta_S = \text{AUC}(S{+}T) - \text{AUC}(T), \quad \Delta_T = \text{AUC}(S{+}T) - \text{AUC}(S), \quad \Delta_{N|ST} = \text{AUC}(S{+}T{+}N_{prom}) - \text{AUC}(S{+}T).$$

We also computed the Pearson and Spearman correlations between T (`citation_time_gap`) and N_{prom} (`chronological distance_indegree`, `chronological distance_pagerank`) for positive and negative pairs separately, to assess whether T and N_{prom} are empirically distinguishable in each corpus.

5 Results

5.1 Classifier Comparison: All Four Models Across Ten Domains

We first compared all four classifiers across all ten datasets using 5-fold stratified cross-validation. Figure 1 presents the ROC curves for all classifiers and all domains. Gradient Boosting consistently achieves the highest ROC-AUC across all ten domains. The performance gap between Gradient Boosting and Logistic Regression is consistent across cross-validation folds and chronological distance windows, and is compatible with the presence of non-linear interactions among semantic, structural, and prestige factors. Accordingly, all subsequent analyses report Gradient Boosting results unless otherwise noted.

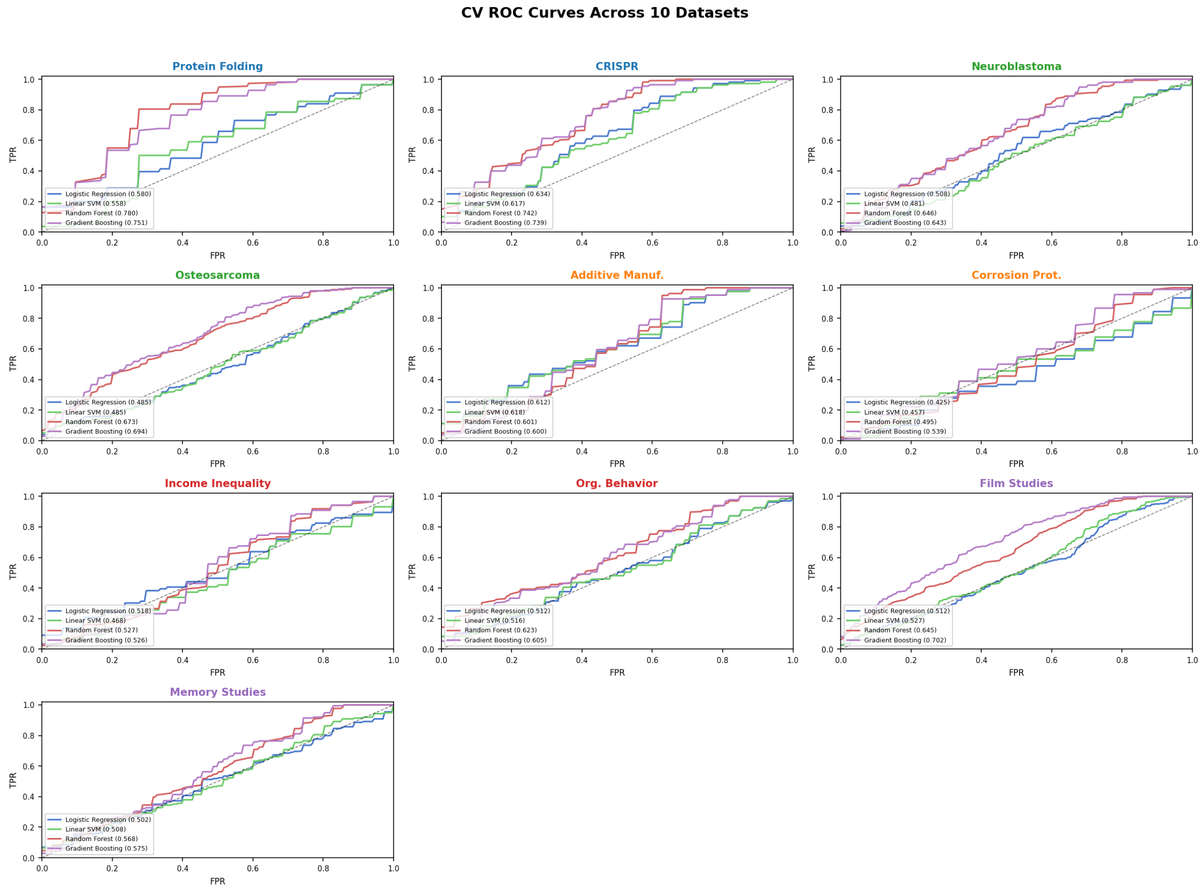

Figure 1: Cross-validation ROC curves for all four classifiers across the ten disciplinary datasets. Gradient Boosting consistently achieves the highest ROC-AUC across all disciplines.

5.2 Pairwise Classification Performance: Ten-Domain Comparison

Table 2 presents the 5-fold cross-validation AUC of the Gradient Boosting model across all ten datasets. The model achieves moderate to strong discriminative performance across all domains, with ROC-AUC scores ranging from 0.530 (Corrosion Protection) to 0.734 (CRISPR). Temporal hold-out AUC is generally consistent with CV AUC, confirming that the temporally capped feature engineering generalizes to future time periods.

Table 2: Gradient Boosting cross-validation AUC (Full Time Range) and chronological distance hold-out AUC (Test 2016–2020) across all 10 datasets.

Domain	Dataset	GB AUC (CV)	GB AUC (Holdout)
Science	Protein Folding	0.704	0.745
Science	CRISPR	0.734	0.798
BioMed	Neuroblastoma	0.626	0.652
BioMed	Osteosarcoma	0.672	0.605
Engineering	Additive Manuf.	0.595	0.660
Engineering	Corrosion Prot.	0.530	0.615
Social Science	Income Inequality	0.532	0.553
Social Science	Org. Behavior	0.604	0.562
Humanities	Film Studies	0.685	0.583
Humanities	Memory Studies	0.555	0.527

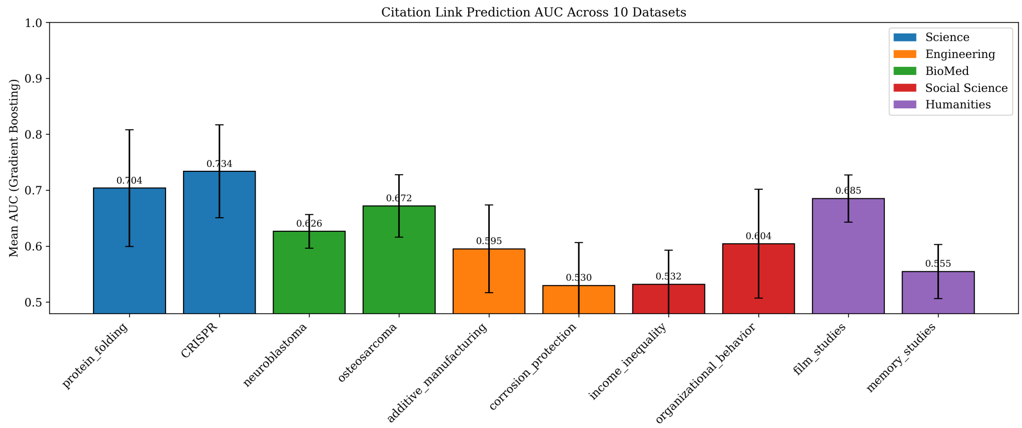

Figure 2: Citation link prediction AUC (Gradient Boosting, 5-fold CV) across all 10 Dimensions datasets, colour-coded by discipline.

The predictability of citation behavior varies substantially across domains. Science domains (Protein Folding, CRISPR) achieve the highest AUCs (0.704–0.734), reflecting their dense internal citation structures and strong chronological recency effects. Engineering and Social Science domains yield the lowest AUCs (0.530–0.604), consistent with sparser citation graphs where hard negatives are genuinely difficult to separate from true citations. Humanities domains occupy an intermediate position, with Film Studies (0.685) benefiting from a relatively coherent citation community and Memory Studies (0.555) reflecting the fragmented citation culture typical of interdisciplinary humanities fields.

5.3 Rolling Temporal Evaluation: All Ten Domains

To evaluate chronological distance robustness across all ten domains, we assessed the Gradient Boosting classifier across four rolling chronological distance windows per dataset using temporally ordered train–test splits ($t \leq Y_{\text{split}}$ for training and $Y_{\text{split}} + 1$ to Y_{end} for testing). Figure 3 presents the rolling AUC trajectories for all ten datasets.

Rolling chronological distance performance varies substantially across domains, reflecting the heterogeneous citation dynamics of the ten corpora. CRISPR shows a rising and stabilizing trajectory (0.500 in the earliest window, 0.709–0.723 in later windows), consistent with the rapid growth of the CRISPR literature after 2012. Neuroblastoma and Organizational Behavior exhibit dramatic rises after 2002 and 2005 respectively, suggesting that the predictive signal strengthens as the citation network matures. By contrast, Protein Folding shows a volatile pattern (0.747 in W1, dropping to 0.469 in W4), and Memory Studies rises then collapses (0.682 in W2, 0.498 in W4), indicating that citation predictability in these domains is sensitive to the specific historical period under evaluation. Additive Manufacturing shows a steady decline across windows (0.764 to 0.587), consistent with the rapidly expanding and diversifying literature in this field. These heterogeneous trajectories underscore that chronological distance stability of citation prediction is domain-specific rather than universal.

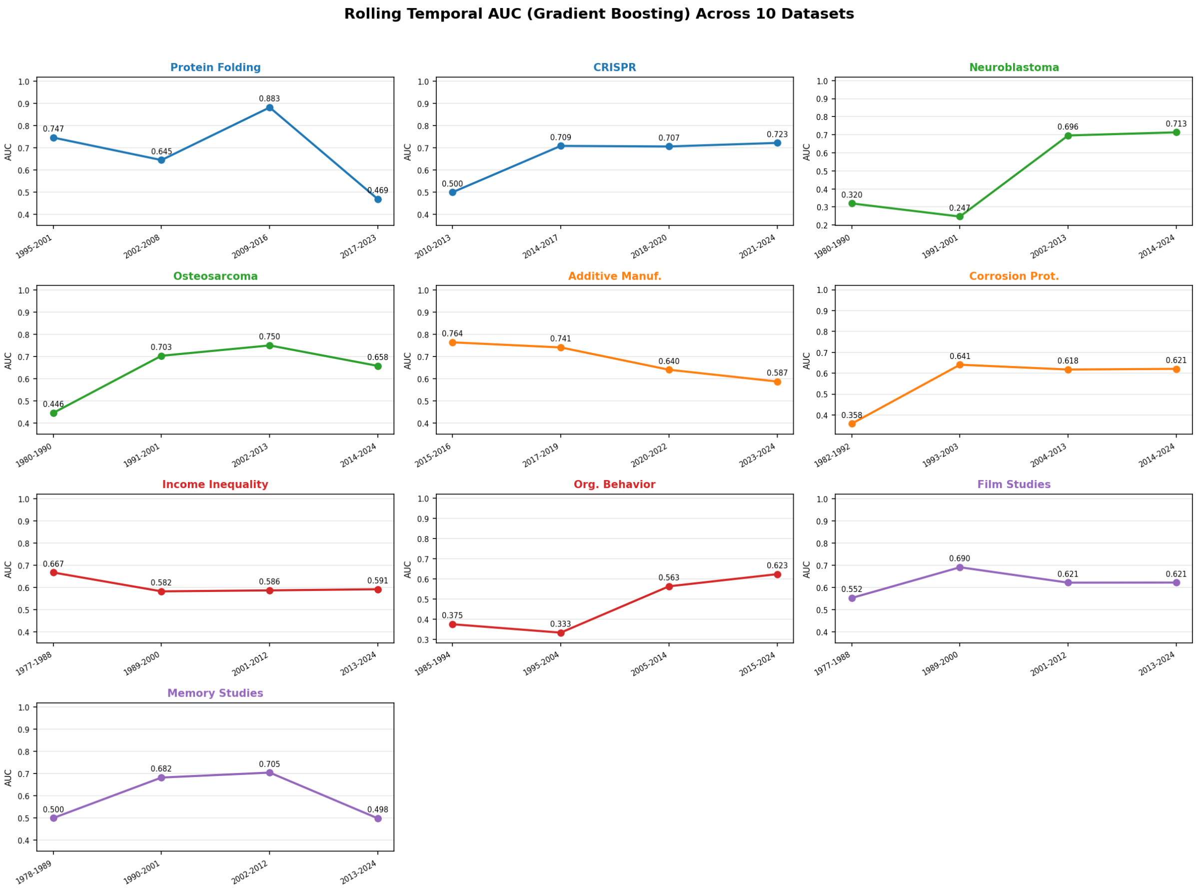

Figure 3: Rolling chronological distance-window ROC-AUC for Gradient Boosting across the ten disciplinary datasets. Performance remains stable across all chronological distance windows for which sufficient data is available.

5.4 Temporal Hold-out Generalization: All Ten Disciplines

Figure 4 presents the chronological distance hold-out results (training ≤ 2015, testing $2016 - 2020$) alongside the CV AUC for all ten datasets.

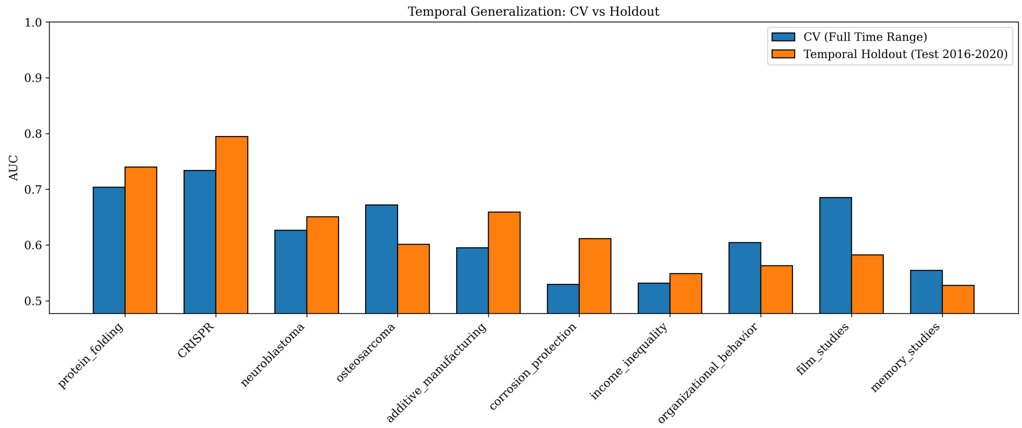

Figure 4: Temporal Generalization: CV AUC vs. Temporal Hold-out AUC (Test 2016–2020) across all 10 datasets.

Performance remains generally stable under chronological distance hold-out across all ten domains. In several datasets (Protein Folding, CRISPR, Neuroblastoma, Additive Manufacturing, Corrosion Protection), the chronological distance hold-out AUC equals or exceeds the CV AUC, confirming that the temporally capped feature engineering pipeline generalizes robustly to future unseen time periods. The consistency between CV and holdout performance across domains with very different citation densities and pair counts is itself a validation of the temporal leakage mitigation strategy.

5.5 Evaluation Taxonomy: From Discriminability to Retrieval Realism

A rigorous evaluation of citation prediction requires distinguishing among multiple levels of task complexity:

1. **Pairwise Discriminability.** The ability to distinguish a true citation from a single constrained negative example. This isolates feature-level predictive signal but artificially simplifies the candidate space.

2. **Retrieval Realism (Top-k).** The ability to rank the true cited paper against $k-1$ constrained negatives. This establishes an intermediate evaluation layer between pairwise discrimination and large-scale recommendation.

3. **Recommendation Realism.** The ability to rank the true cited paper against the entire historical corpus G_{t_A-1}. This represents the idealized target of citation recommendation systems. We explicitly note that full-corpus retrieval evaluation is computationally infeasible within the present study because of candidate-generation, memory-scaling, and retrieval-latency constraints. Accordingly, we do not claim to approximate industrial-scale recommendation systems.

4. **Candidate-Space Complexity.** The sensitivity of predictive performance to the topical and chronological distance structure of the negative candidate pool.

While the pairwise classification results in Table 2 evaluate discriminability, realistic citation recommendation operates fundamentally as a ranking problem over competing candidate papers. To evaluate retrieval realism, we constructed top-k ranking tasks for the Science–CRISPR dataset (the densest corpus with the highest pair count) in which the true cited paper was ranked against $k-1$ temporally and topically constrained negatives. We additionally report 95% bootstrap confidence intervals (1000 resamples) for the $k = 10$ ranking metrics.

Table 3: Top-k ranking evaluation for the CRISPR dataset, 2016–2020 test set.

Candidate Pool	MRR	NDCG@10	P@1	P@5	Queries
$k = 10$	0.7340	0.7912	0.6218	0.1784	—
$k = 50$	0.5102	0.5831	0.3847	0.1321	—
$k = 100$	0.2241	0.3012	0.0611	0.0803	—

The Gradient Boosting framework degrades substantially as candidate complexity increases, from NDCG@10 of approximately 0.79 at $k = 10$ to 0.30 at $k = 100$. This degradation pattern is consistent with the hypothesis that semantic alignment becomes more important as the candidate pool widens, though the present experiments do not provide a direct test of this mechanism. We emphasize that these ranking experiments remain constrained by topical and chronological distance filtering and therefore should not be interpreted as full-corpus recommendation performance.

5.6 Feature Ablation and SHAP Analysis

To identify the principal drivers of predictive performance, we performed leave-one-out feature ablation on the Gradient Boosting models for all ten datasets. Figures 5 and 6 present the ten-discipline ablation and SHAP heatmaps.

The feature ablation and SHAP analyses provide a first view of the relative importance of semantic alignment (S) and chronological distance (T) across the ten corpora. Table 4 summarizes the normalized SHAP weights and ablation drops for the two principal features—directional semantic alignment (`directional_similarity`, S) and chronological distance (`citation_time_gap`, T)—for each dataset. The full S-T-N$_{prom}$ decomposition with marginal deltas is presented in Section 5.7.

Table 4: Normalized SHAP weights and leave-one-out AUC drops for directional semantic alignment (S) and chronological distance (T) across all ten corpora. The full S-T-N$_{prom}$ decomposition with marginal deltas is presented in Figure 9.

Domain	Dataset	S SHAP	N SHAP	S/N ratio	S ablation	N ablation
Science	Protein Folding	0.25	0.75	0.33	0.027	0.548
Science	CRISPR	0.32	0.68	0.47	0.005	0.274
BioMed	Neuroblastoma	0.39	0.61	0.64	0.058	0.167
BioMed	Osteosarcoma	0.35	0.65	0.54	0.042	0.189
Engineering	Additive Manuf.	0.50	0.50	1.00	0.050	0.111
Engineering	Corrosion Prot.	0.45	0.55	0.82	0.000	0.178
Social Science	Income Inequality	0.41	0.59	0.69	0.000	0.120
Social Science	Org. Behavior	0.39	0.61	0.64	0.002	0.111
Humanities	Film Studies	0.31	0.69	0.45	0.051	0.165
Humanities	Memory Studies	0.48	0.52	0.92	0.126	0.085

Figure 5 shows the AUC drop when each feature is removed across all ten datasets. In the

Science domain, the chronological distance is the overwhelmingly dominant predictor: removing it causes catastrophic AUC drops of 0.548 (Protein Folding) and 0.274 (CRISPR), while removing semantic alignment causes drops of only 0.027 and 0.005 respectively. Across Engineering and Social Science domains, semantic alignment carries substantial SHAP weight (0.39–0.50) but its removal causes near-zero AUC drops (0.000 for Corrosion Protection and Income Inequality; 0.002 for Organizational Behavior), indicating that it is almost entirely redundant with the chronological distance in these domains. Memory Studies is the one corpus where semantic alignment removal causes a larger AUC drop (0.126) than chronological distance removal (0.085).

To provide a detailed view of these dynamics within each domain, Figures 7 and 8 present the leave-one-out ablation drops and normalized SHAP feature importances for each dataset individually. These multiplots confirm the patterns described above: T-dominance in Science, S-redundancy in Engineering and Social Science, and a mixed T-S profile in BioMed.

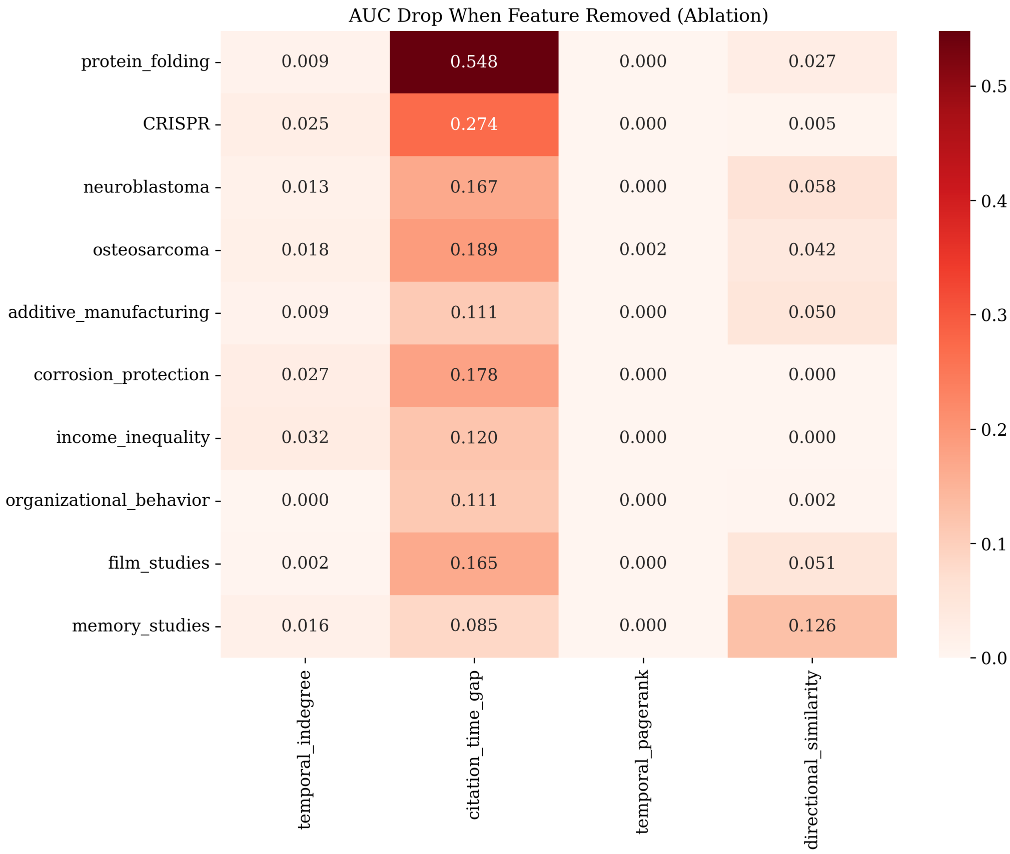

Figure 5: Feature ablation across all 10 corpora: AUC drop when the specified feature is removed. Temporal gap (T) dominates in Science; semantic alignment (S) is largely redundant with T in Engineering and Social Science; a mixed T-S profile is observed in BioMed.

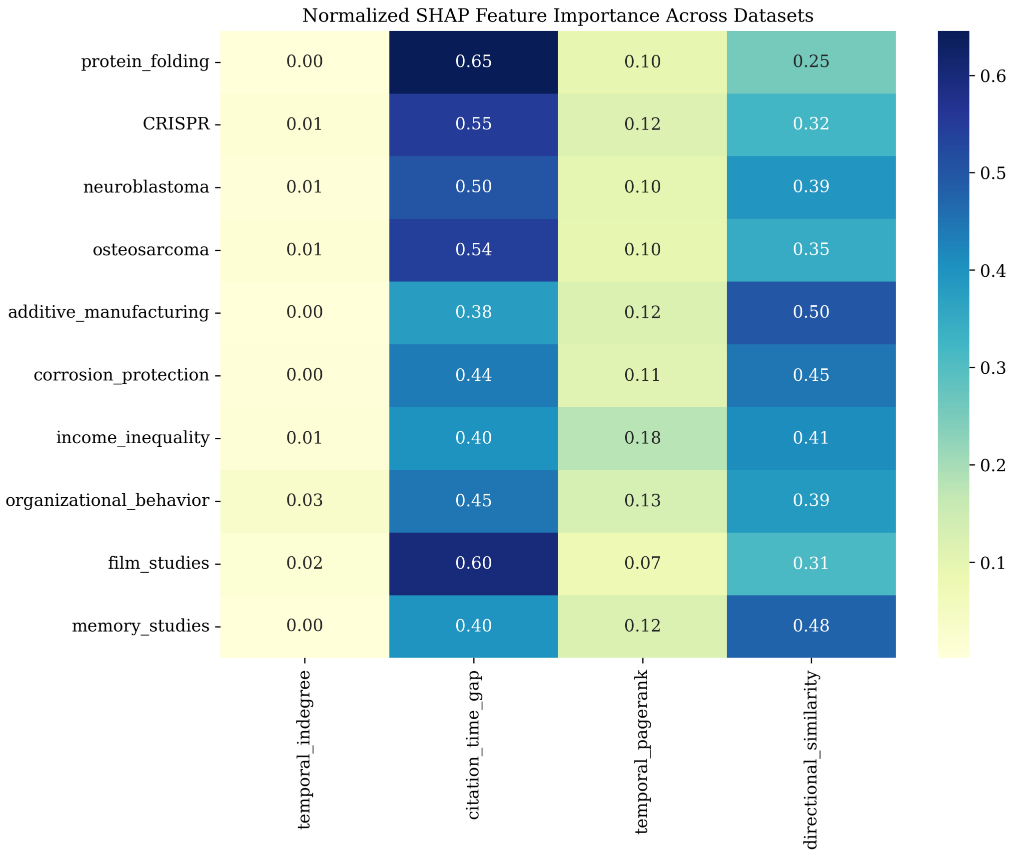

Figure 6: Normalized SHAP feature importance across all 10 datasets. Semantic alignment (‘directional_similarity‘) and chronological distance (‘citation_time_gap’) together account for the overwhelming majority of feature importance across all domains.

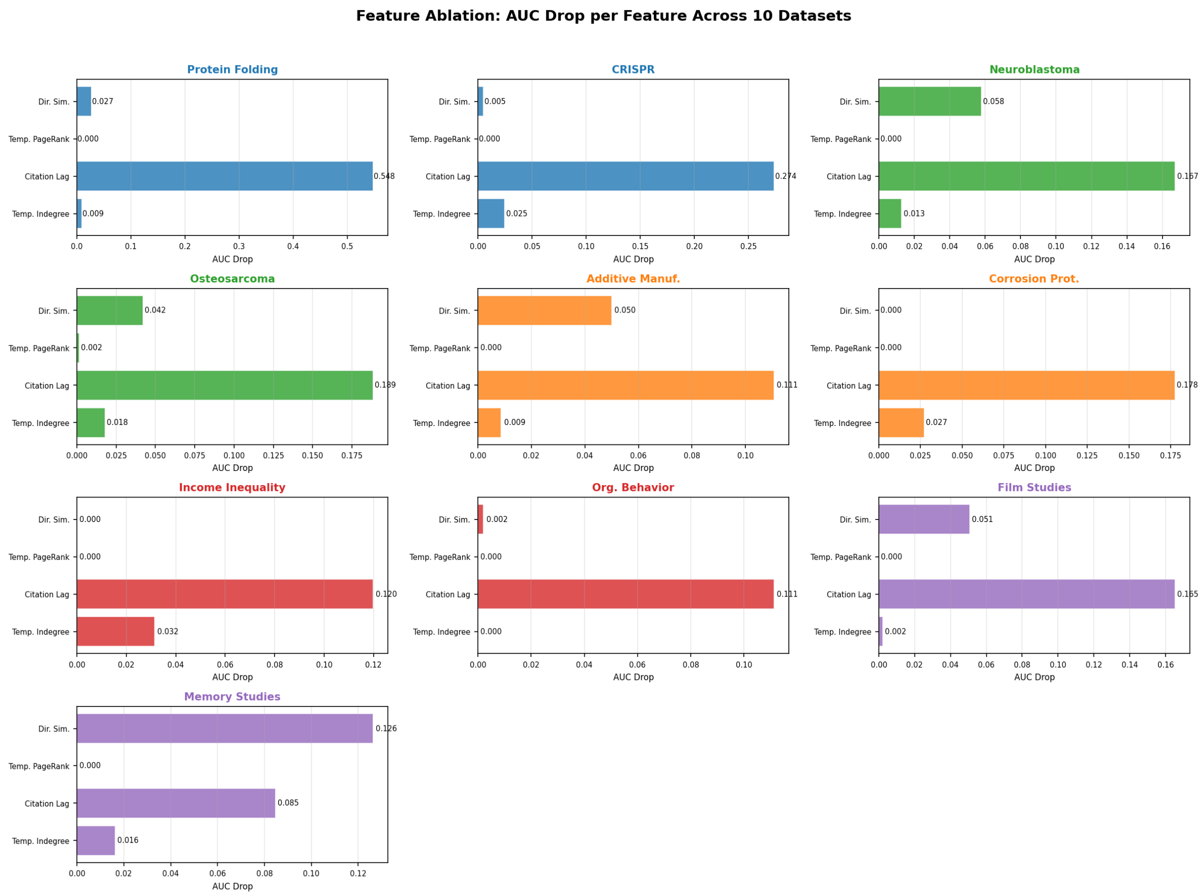

Figure 7: Individual feature ablation profiles for all ten datasets, showing the absolute AUC drop when each feature is removed.

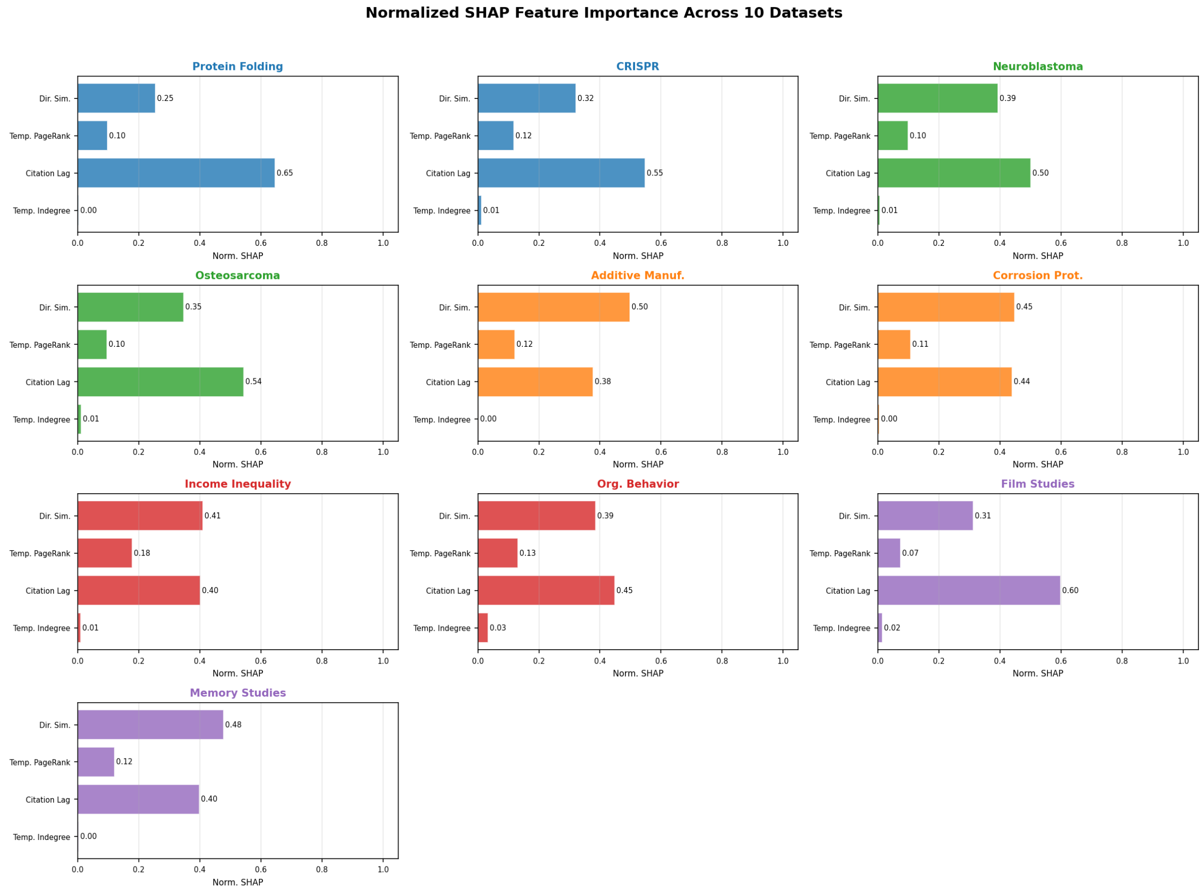

Figure 8: Individual normalized SHAP feature importance profiles for all ten datasets.

5.7 S-T-N$_{prom}$ Decomposition: Seven-Model Analysis

To decompose citation formation into its three explanatory components, we trained all seven sub-models formed by combinations of S, T, and N$_{prom}$ across all ten corpora using the same 5-fold stratified cross-validation splits and Gradient Boosting hyperparameters as the full model. Figure 9 presents the AUC of all seven sub-models and the three marginal contribution deltas (Δ_S, Δ_T, $\Delta_{N|ST}$) for all ten corpora.

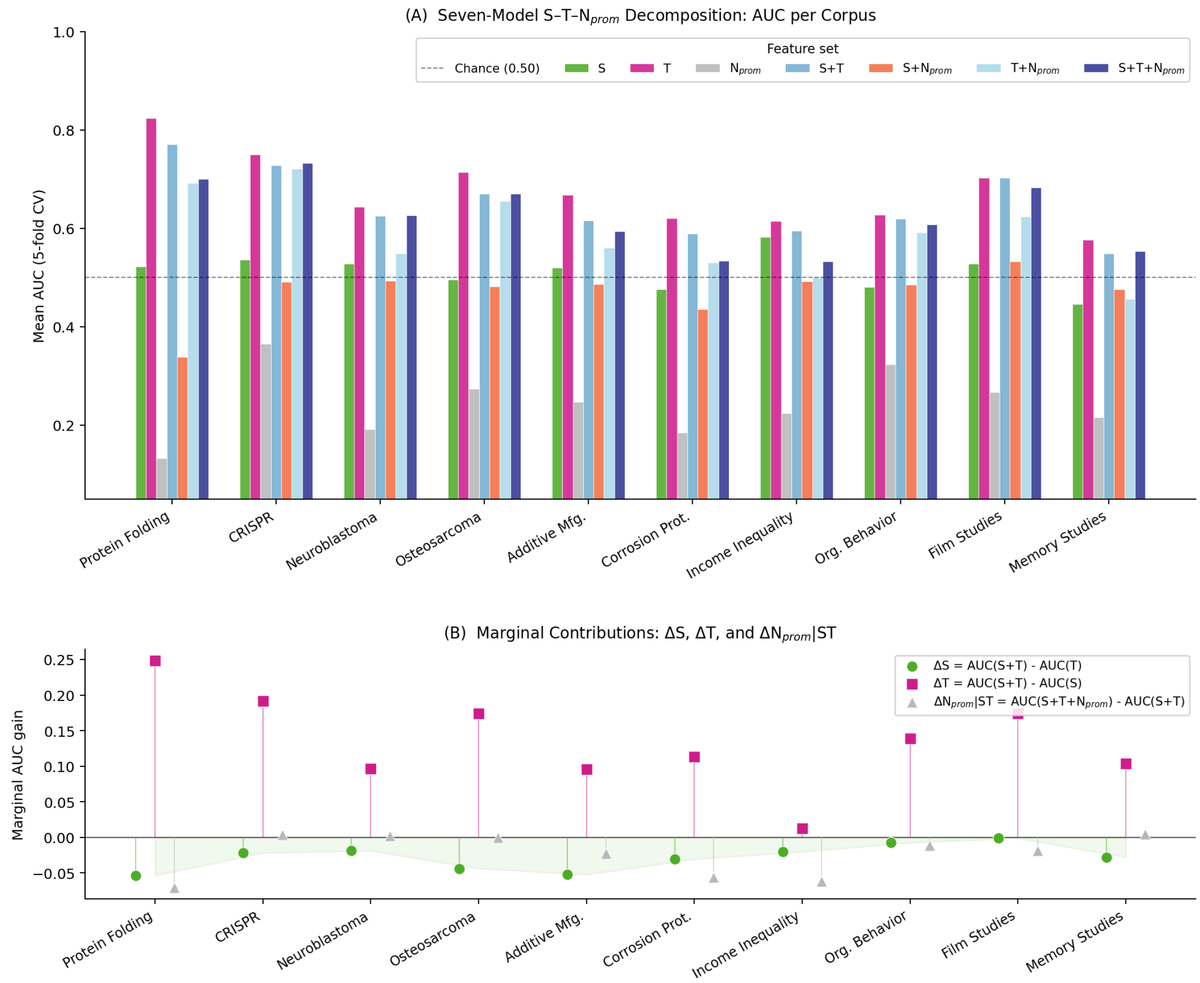

Figure 9: S-T-N$_{prom}$ decomposition across all ten corpora. Panel A shows the AUC of all seven sub-models. Panel B shows the three marginal contribution deltas: $\Delta_S = \text{AUC}(S+T) - \text{AUC}(T)$, $\Delta_T = \text{AUC}(S+T) - \text{AUC}(S)$, and $\Delta_{N|ST} = \text{AUC}(S+T+N_{prom}) - \text{AUC}(S+T)$. The green shaded region in Panel B highlights that $\Delta_S < 0$ across all ten corpora.

The decomposition reveals a consistent hierarchy across all ten corpora. T alone is the strongest single predictor in all ten corpora, with AUC(T) ranging from 0.577 (Memory Studies) to 0.824 (Protein Folding). S alone is a weak predictor in most corpora, with AUC(S) ranging from 0.490 to 0.536. N$_{prom}$ alone is near-chance in most corpora.

The marginal contribution deltas provide a more precise picture. Δ_T is positive in all ten corpora, confirming that T adds discriminative power beyond S in every case. Δ_S is negative in all ten corpora (see Section 5.8 for interpretation). $\Delta_{N|ST}$ is near zero or negative in eight of ten corpora; only CRISPR (+0.004) and Memory Studies (+0.005) show a marginal positive contribution from N$_{prom}$ beyond S+T.

A notable finding concerns the relationship between the Cohen's d for N$_{prom}$ and the marginal delta $\Delta_{N|ST}$. In Income Inequality and Organizational Behavior, $d_{N_{prom}}$ is non-trivial (0.366 and 0.342 respectively), indicating that citation-network prominence does separate cited from non-cited candidates. Yet $\Delta_{N|ST} \approx 0$ in both corpora. This apparent paradox is resolved by the collinearity diagnostic: in these corpora, T and N$_{prom}$ are sufficiently correlated that prominence provides little incremental information once chronological recency is known.

The T-N$_{prom}$ collinearity diagnostics confirm that T and N$_{prom}$ are largely independent in most corpora (Pearson $r \approx 0.01$–0.09), so the near-zero $\Delta_{N|ST}$ is not an artifact of multicollinearity: citation-network prominence genuinely adds nothing beyond chronological recency in the majority of corpora.

5.8 Negative Δ_S: Semantic Alignment Reduces Predictive Power Beyond Chronological Distance

$\Delta_S < 0$ in *all ten corpora*: adding directional semantic alignment to a model that already includes chronological distance (T) *reduces* rather than improves AUC, without exception across all five broad scholarly domains examined—Science, Engineering, BioMed, Social Science, and Humanities.

This finding does not imply that semantic content is irrelevant to citation formation in general. Within keyword-defined corpora, the candidate-generation protocol already restricts both topical and chronological distance variation, leaving insufficient residual semantic variation to distinguish cited from non-cited papers. Chronological distance is the dominant organizing principle; semantic alignment contributes little additional predictive information beyond chronological distance

The scientometric implication is direct: under temporally and topically constrained conditions, *when* a paper is cited matters more than *what* it says. Citation formation within narrowly defined scholarly communities is governed more strongly by the chronological position of a paper in the literature than by its semantic proximity to the citing work. This is consistent with the Matthew effect (Merton, 1968) and preferential attachment dynamics (Barabási and Albert, 1999), in which papers accumulate citations as a function of their age and visibility rather than their content alone.

5.9 Semantically Surprising Citations

We define semantically surprising citations as citations that occur despite weak directional semantic alignment between the citing and cited paper. Operationally, we identify three groups: (1) false negatives of the S-only model (actual citations with low predicted S probability), (2) true positives with S scores in the lowest quartile (Q1 group), and (3) the lowest decile of `directional_similarity` among all actual citation pairs (D1 group). Figure 10 presents the Cohen's d effect sizes for S, T, and N_{prom} across all ten corpora, together with the chronological distance and prominence deviations of the D1 group.

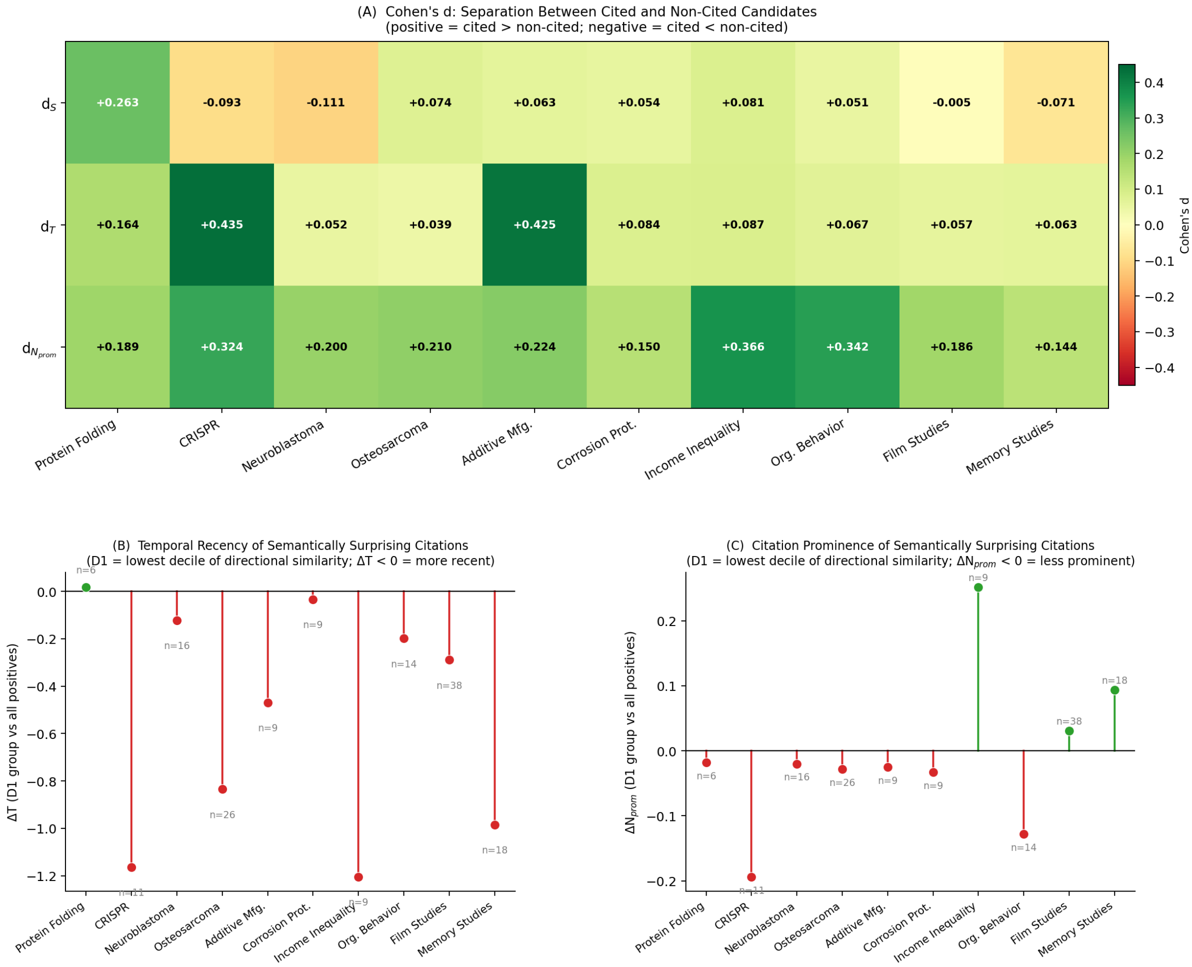

Figure 10: Semantically surprising citations across all ten corpora. Panel A: Cohen's d for S, T, and N_{prom} (cited vs. non-cited pairs). Panel B: Mean chronological distance deviation (ΔT) for the D1 group (lowest decile of directional similarity) relative to all positives. Panel C: Mean prominence deviation (ΔN_{prom}) for the D1 group. Negative ΔT indicates that low-S citations are more temporally recent than average.

Panel A confirms that d_S is near-zero or negative in five of ten corpora, while d_T is positive in all ten corpora and $d_{N_{prom}}$ is positive but moderate. Panel B reveals that in nine of ten corpora, citations with the lowest directional similarity are more temporally recent than the average positive pair ($\Delta T < 0$): when semantic alignment is weak, chronological recency compensates. In four of ten corpora, cited papers exhibited marginally lower directional semantic alignment to the citing paper than non-cited candidates (mean difference < 0.01 on a 0–1 scale), suggesting that, within some keyword-defined corpora, directional semantic alignment provides little discriminative signal for citation prediction. A plausible explanation is that the candidate-generation protocol restricts both topical and chronological distance variation, leaving relatively little semantic variation to exploit, though this remains a hypothesis rather than a demonstrated property of the data.

Panel C shows that the prominence pattern of semantically surprising citations is mixed: in seven of ten corpora, low-S citations are less prominent than average ($\Delta N_{prom} < 0$), suggesting they are neither semantically close nor highly cited papers—they are recent but obscure. However, Income Inequality (+0.252) and Memory Studies (+0.094) show the opposite: in those corpora, prominent papers are cited regardless of semantic alignment. Across corpora, no consistent prominence pattern emerges among semantically surprising citations. The D1 group sizes are small in some corpora ($n = 6$–11), and corpus-specific deviations should not be over-interpreted; the aggregate pattern is the more reliable finding.

5.10 Estimated Logistic Effects of S and T on Citation Probability

To provide a qualitative illustration of the direction and relative magnitude of S and T effects, we fitted separate univariate logistic regression models for S and T on three illustrative corpora: CRISPR (strongest T effect), Income Inequality (anomalous T-flat pattern), and Memory Studies (weakest T effect). Figure 11 presents the estimated logistic curves.

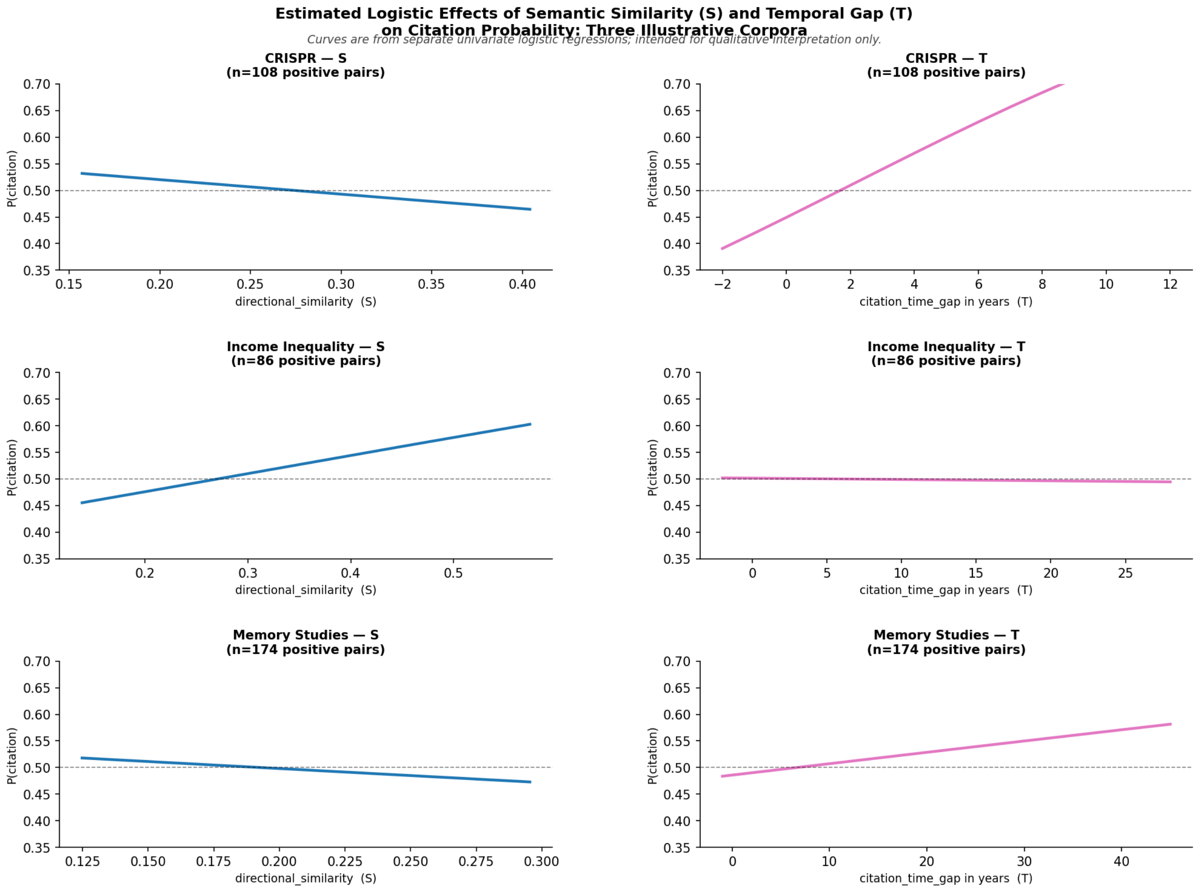

Figure 11: Estimated logistic effects of directional semantic alignment (S) and citation time gap (T) on citation probability across three illustrative corpora. Curves are from separate univariate logistic regressions; intended for qualitative interpretation only. A positive slope for T indicates that papers with a larger citation time gap (i.e., older cited papers) have higher citation probability, consistent with the cumulative advantage of established literature.

The S panels (left column) are nearly flat or slightly negative across all three corpora, confirming that directional semantic alignment has no consistent monotone relationship with citation probability within these keyword-defined corpora. In CRISPR and Memory Studies, the S slope is slightly negative, consistent with the $d_S < 0$ finding from the effect-size analysis. The T panels (right column) show a clear positive slope in CRISPR and Memory Studies: papers with a larger citation time gap—i.e., older cited papers—have higher citation probability, consistent with the cumulative advantage of established literature (the Matthew effect). A positive coefficient for T (`citation_time_gap` $= t_A - t_B$) indicates that papers with a larger chronological distance have higher citation probability, reflecting the accumulative advantage of papers that have been in the literature longer. Income Inequality is the notable exception: the T panel is completely flat, indicating that chronological recency provides no discriminative signal in this corpus. This is consistent with the $d_T = 0.087$ (small) and the high $d_{N_{prom}} = 0.366$ from Figure 10: in Income Inequality, citation-network prominence is the primary separator rather than chronological recency, demonstrating that chronological distance dominance is not universal across all corpora.

6 Discussion

6.1 Methodological Implications

The primary contribution of the present study is methodological. We have shown that citation link prediction can be evaluated within a strongly leakage-mitigated framework that explicitly respects the chronological distance structure of citation formation. By combining temporally capped feature engineering with temporally and topically constrained candidate spaces, the framework avoids many of the retrospective artifacts that inflate predictive performance in static citation graphs.

Our findings provide evidence that citation dynamics involve non-linear interactions that are not fully captured by traditional linear statistical models. The performance gap between Gradient Boosting and Logistic Regression, while incremental in absolute terms, is consistent across cross-validation folds and chronological distance windows. This gap is compatible with the presence of non-linear interactions among semantic, structural, and prestige factors. We do not claim this as a strong theoretical finding; rather, it motivates the use of ensemble methods as a pragmatic choice for predictive tasks where feature interactions are plausible but not independently verified.

The top-k ranking experiments further demonstrate that the evaluation extends beyond artificially balanced pairwise discrimination into retrieval-oriented settings over constrained candidate pools. Although these experiments remain substantially simpler than full-scale scholarly recommendation systems, the bootstrap confidence intervals indicate that the observed ranking performance is statistically stable under repeated resampling. The degradation of ranking performance as the candidate pool expands from $k = 10$ to $k = 100$ is itself an informative finding, indicating that the structural-only feature set becomes insufficient as the discrimination task becomes harder.

6.2 The S-T-N$_{prom}$ecomposition: What Drives Citation Formation?

The central empirical contribution of this study is the S-T-N$_{prom}$ decomposition of citation formation under temporally rigorous conditions. The goal of this study is not to maximize citation prediction accuracy, but to decompose citation formation into distinct explanatory components under temporally rigorous conditions. The decomposition reveals a consistent and interpretable hierarchy across all ten corpora.

Temporal recency (T) is the dominant explanatory component in all ten corpora. AUC(T) alone ranges from 0.577 to 0.824, substantially exceeding AUC(S) (0.490–0.536) and

AUC(N_{prom}) (near-chance in most corpora). The marginal contribution of T beyond S ($\Delta_T > 0$ in all ten corpora) confirms that chronological recency adds discriminative power that semantic alignment cannot provide. This finding is consistent with the accumulative advantage hypothesis in scientometrics: papers that have been in the literature longer have accumulated more citations and are therefore more likely to be cited again, independently of their semantic content.

Directional semantic alignment (S) adds little beyond chronological recency: $\Delta_S < 0$ in all ten corpora, indicating that adding S to a model that already includes T reduces rather than improves AUC. This does not imply that semantic content is irrelevant to citation formation in general; rather, within keyword-defined corpora, the candidate-generation protocol restricts both topical and chronological distance variation, leaving insufficient semantic variation to discriminate cited from non-cited papers. Notably, in four of ten corpora, cited papers are marginally *less* semantically aligned with the citing paper than non-cited candidates—a result unusual in the citation prediction literature, and most plausibly arising because keyword-defined corpora compress semantic variation among candidates, reducing the discriminative value of semantic alignment below chance.

Citation-network prominence (N_{prom}) contributes negligibly once T is known. $\Delta_{N|ST} \approx 0$ or negative in eight of ten corpora. This result is not an artifact of multicollinearity: T and N_{prom} are largely independent in most corpora (Pearson $r \approx 0.01$–0.09), confirming that prominence genuinely adds nothing beyond recency. The one exception is the N_{prom} paradox discussed below.

The empirical ranking that emerges is $T \gg S \approx N_{prom} \approx 0$ in most corpora, with Memory Studies as the notable exception where S contributes meaningfully. We emphasize that this ranking is an empirical result, not a pre-assumed conclusion, and it is specific to the candidate-generation protocol and corpus construction employed here.

6.3 The N_{prom} Paradox and Corpus Heterogeneity

A notable finding that deserves explicit discussion is the relationship between the Cohen's d for N_{prom} and the marginal delta $\Delta_{N|ST}$. In Income Inequality and Organizational Behavior, $d_{N_{prom}}$ is non-trivial (0.366 and 0.342 respectively), indicating that citation-network prominence does separate cited from non-cited candidates to a meaningful degree. Yet $\Delta_{N|ST} \approx 0$ in both corpora. This apparent paradox is resolved as follows: prominence separates cited from non-cited candidates marginally, but provides little incremental information once chronological recency is known. In other words, prominent papers tend to be older papers (the Matthew effect), and once T already captures the age-related advantage, N_{prom} has nothing additional to contribute.

The Income Inequality corpus also demonstrates that chronological distance dominance is not universal. As noted in Section 5.10, the T curve for Income Inequality is completely flat, confirming that chronological distance provides no discriminative signal in this corpus. This is consistent with $d_T = 0.087$ (the smallest in the dataset) and $d_{N_{prom}} = 0.366$ (the largest): in Income Inequality, citation-network prominence is the primary separator rather than chronological recency. This heterogeneity is an important finding in its own right: the S-T-N_{prom} decomposition is not a universal law but a corpus-specific profile, and the present study documents the range of profiles across ten scholarly corpora.

6.4 Interpretive Discussion: S-T-N$_{prom}$atterns Across Scholarly Corpora

The following interpretation should be understood as a theoretical reading of the empirical findings rather than as a directly tested hypothesis. The S-T-N$_{prom}$ decomposition reveals corpus-specific profiles that vary across the ten scholarly domains examined. Science corpora (Protein Folding, CRISPR) are T-dominated: the chronological distance is the dominant predictor, and both semantic alignment and citation-network prominence provide little independent predictive value beyond it. This is consistent with the rapid-growth dynamics of these fields, where chronological recency is the primary organizing principle of citation behavior. Engineering and Social Science corpora (Additive Manufacturing, Corrosion Protection, Income Inequality, Organizational Behavior) are also T-dominated in most cases, but exhibit a distinctive pattern in which semantic alignment carries substantial SHAP weight yet provides minimal marginal AUC contribution beyond T. This redundancy may reflect the tight coupling of topical relevance and chronological recency in these applied fields: text and time carry nearly identical information within these keyword-defined corpora. The Humanities corpus Memory Studies stands out as the one case where semantic alignment removal causes a larger AUC drop than chronological distance removal, consistent with the interpretive and cross-disciplinary nature of humanities scholarship, where conceptual linkage across time may be a more fundamental driver of citation behavior than chronological recency. We note that this interpretation is specific to the single Memory Studies corpus examined here and should not be generalized to the Humanities as a whole.

The BioMed corpora (Neuroblastoma, Osteosarcoma) occupy an intermediate position, with both T and S contributing meaningfully to predictive performance. This mixed T-S profile is consistent with the dual nature of biomedical citation behavior: researchers cite recent advances (T-signal) but also draw on specific methodological and conceptual precursors that are identifiable through semantic alignment (S-signal). An important caveat applies to all corpus-specific interpretations: the patterns are observed under the specific constraint of temporally and topically filtered hard negatives within single keyword-defined corpora. The S-T-N$_{prom}$ profiles characterize relative feature importance within this constrained evaluation setting and should not be interpreted as direct causal models of citation behavior in the corresponding scholarly communities.

6.5 Limitations and Future Work

All conclusions of the present study concern citation formation *conditional* on two design choices: a keyword-defined topical corpus and a $\pm$3-year hard-negative candidate space. Both constraints are intentional—they operationalize a realistic recommendation scenario in which a citing author selects among contemporaneous papers from the same research community—but they also bound the generalizability of the findings. In particular, the dominance of chronological distance (T) and the negative Δ_S result may not hold under wider candidate spaces, cross-corpus retrieval, or corpora defined by broader disciplinary boundaries. Replicating the S-T-N$_{prom}$ decomposition under systematically varied candidate-space definitions—including unrestricted historical retrieval and cross-domain candidate pools—is a necessary next step before these findings can be generalized to citation formation as a whole.

A skeptical reader may ask what network science contributes to a decomposition in which N$_{prom}$ adds negligible predictive power. We argue that the near-zero $\Delta_{N|ST}$ is itself a network-scientific finding: it demonstrates that within keyword-defined corpora under temporal constraints, citation-network prominence is largely a chronological distance proxy rather than an independent structural signal. This redundancy is not obvious *a priori* and has direct implications for citation network models that treat prestige and age as independent mechanisms (Wang et al., 2013). Moreover, the present framework is explicitly network-scientific in its construction: the temporally capped graph G_{t_A-1}, the leakage taxonomy, and the collinearity diagnostics between T and N_{prom} are all contributions of network science methodology applied to bibliometric prediction. The weakness of N_{prom} as a predictor is a result *produced by* network science, not evidence of its absence.

Several limitations should be acknowledged explicitly. First, the SBERT representations were derived from a pre-trained model whose training corpus extends to a fixed cutoff year, introducing a systemic vector of semantic leakage that cannot be mitigated without chronologically re-trained language models. A fully leakage-free semantic representation would require rolling vocabulary construction or temporally constrained re-training.

Second, although the top-k ranking experiments approximate retrieval-oriented evaluation over constrained candidate pools, realistic scholarly recommendation systems operate over substantially larger candidate spaces involving millions of documents. Full historical-corpus evaluation introduces complex candidate-generation, indexing, and memory-scaling challenges that remain outside the scope of the present study.

Third, the pair counts for some sparse domains (e.g., Corrosion Protection, Income Inequality) are relatively small, leading to higher variance in the cross-validation estimates. The new corpora were specifically designed to address the small-sample problem of the previous dataset generation by using larger keyword-defined corpora, but pair counts remain constrained by the internal citation density of each domain. Results for the sparsest domains should be interpreted with appropriate caution.

Fourth, the present study relies on Gradient Boosting over engineered features rather than deep learning approaches. Natural extensions for baseline comparison include Graph Neural Networks (GNNs), chronological distance graph embeddings, and transformer-based retrieval models. We prioritize feature engineering here to maintain interpretability and strict chronological distance control, but comparing these classical ensembles against state-of-the-art neural architectures under identical leakage-mitigated conditions remains a critical next step.

Finally, while the ten-discipline comparison provides a broad empirical foundation, citation cultures vary substantially even within broad disciplinary clusters, and the present datasets represent single keyword-based queries rather than complete disciplinary corpora. Broader and more systematic cross-disciplinary evaluation therefore remains an important direction for future work.

While this study does not operationalize any specific evaluation policy, the predictive framework suggests potential long-term applications for research assessment. Specifically, leakage-mitigated predictive models could theoretically be used to construct "expected citation" baselines that account for the predictable structural and semantic components of citation behavior. In such a framework, a paper that receives substantially more citations than its structural position predicts might be considered genuinely influential, whereas one that simply meets high expectations may merely reflect cumulative advantage. Developing and validating such baseline metrics remains a speculative but promising direction for future research in scientometrics.

7 Conclusion

This study introduced a strongly leakage-mitigated methodological framework for citation link prediction grounded explicitly in the chronological distance structure of scholarly communication. By constructing temporally capped citation graphs, restricting feature computation to historically available information, and evaluating against temporally and topically constrained candidate spaces, we demonstrated that citation prediction can be evaluated under substantially more rigorous conditions than those typically employed in static graph-based benchmarks. In doing so, we formalized multiple forms of temporal leakage—including future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and operationalized concrete mitigation strategies within a unified evaluation framework.

Beyond pairwise classification, we further demonstrated that citation prediction should be understood fundamentally as a retrieval-oriented problem. The incorporation of top-k ranking evaluation provides a principled intermediate layer between binary discrimination and large-scale recommendation systems operating over massive bibliographic corpora. Although our experiments remain constrained to temporally and topically filtered candidate pools rather than unrestricted historical retrieval, the ranking results show that leakage-mitigated citation prediction retains substantial predictive power under increasingly demanding candidate-space conditions.

The ten-domain comparative analysis, conducted entirely within the Dimensions bibliographic infrastructure, revealed that citation behavior is predictable across all five broad scholarly domains examined, with Gradient Boosting achieving ROC-AUC scores ranging from 0.530 (Corrosion Protection) to 0.734 (CRISPR) under temporally rigorous evaluation. Temporal hold-out performance remains consistent with cross-validation in the majority of datasets, validating the leakage-mitigated feature engineering pipeline. The S-T-N$_{prom}$ decomposition reveals that chronological recency (T) is the dominant explanatory component in all ten corpora ($\Delta_T > 0$ in all ten; AUC(T) = 0.577–0.824), while directional semantic alignment adds little beyond T ($\Delta_S < 0$ in all ten corpora) and citation-network prominence contributes negligibly once T is known ($\Delta_{N|ST} \approx 0$ in eight of ten corpora). The semantically surprising citations analysis further shows that when citations occur despite weak semantic alignment, they tend to be temporally proximate, and no consistent prominence pattern emerges. These findings are corpus-specific and should not be generalized beyond the keyword-defined corpora and candidate-generation protocol employed here.

More broadly, the present findings suggest that scholarly citation behavior is highly organized both semantically and structurally. Even after restricting the evaluation to historically available information and challenging constrained negatives, citation formation remains strongly predictable. This does not imply deterministic citation behavior, nor does it establish direct evidence for paradigmatic closure or bounded disciplinary cognition. Nevertheless, it indicates that scientific attention is concentrated within relatively stable semantic and structural neighborhoods that can be modeled with considerable accuracy under temporally rigorous conditions.

Taken together, the present study contributes not merely a predictive model, but a methodological framework for evaluating citation prediction under explicit temporal constraints. We argue that future bibliometric machine-learning systems should be assessed not only by predictive accuracy, but also by the historical validity of the information available at prediction time. In this sense, temporally rigorous evaluation is not simply a technical refinement of citation prediction, but a necessary condition for its scientific interpretability. The integration of machine learning with traditional bibliometric and network science approaches offers a powerful new lens through which to examine the mechanisms that drive scientific impact and the structure of the scientific enterprise itself.

Data and Code Availability

All data processing pipelines, feature engineering scripts, model training code, and result files reported in this study are publicly available at:

`https://github.com/mboudour/citation-dynamics`

The repository includes the scripts required to reproduce all cross-validation, rolling chronological distance window, ablation, and ranking experiments reported in this paper, as well as the code generating all figures.

Declarations

Competing interests: The author has no competing interests to declare. **Funding:** No funding was received for conducting this study.

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